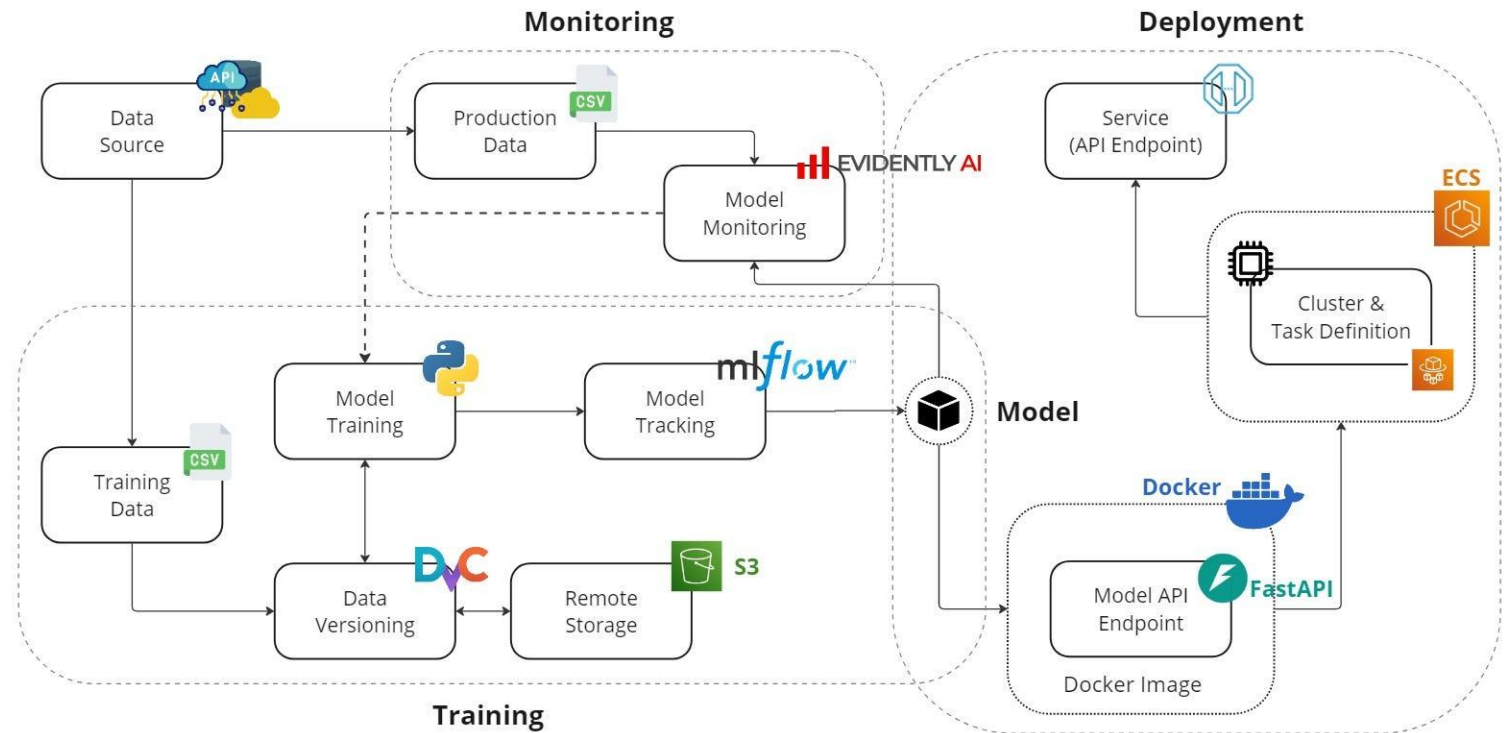


Module 5 – Production Deployment & MLOps

- **Goal:** Packaging and scaling AI systems for the real world.
- **The Problem:** AI environments are fragile. Differences in Python versions, libraries, or GPU drivers can crash a system during deployment.
- **The Solution: Containerization.** We wrap the entire stack—API, Agent logic, and Vector Database—into a single, immutable image.

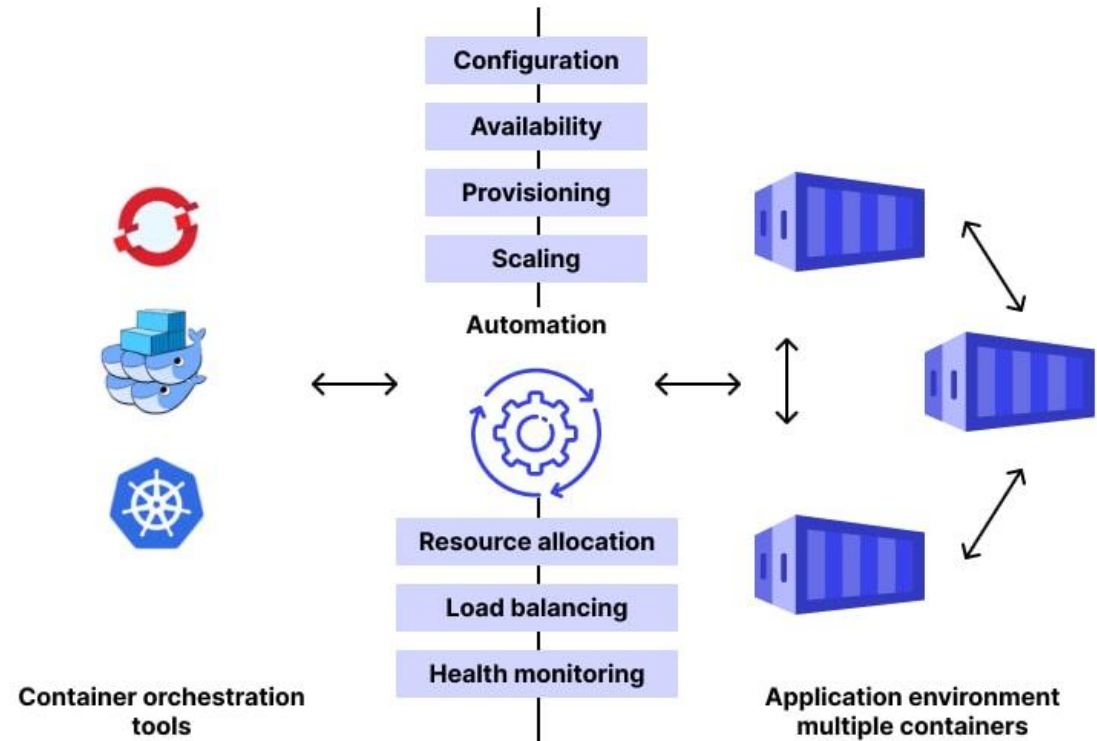
MLOps For Beginners



by Prasad Mahamulkar

Multi-Container Orchestration

- **Docker Compose:** Managing the "Symphony." In production, your Agent API needs to talk to a Vector DB (like Qdrant or FAISS-Server).
- **Networking:** Docker Compose creates a private virtual network where these services communicate securely without exposing every port to the public internet.



Quantization & Observability

- **Quantization (GGUF/AWQ):** Compressing Large Language Models by reducing the precision of weights (e.g., 16-bit to 4-bit) to enable low-latency hosting on standard hardware.
- **Observability:** Using "Traces" to see exactly which step an agent took before it failed. If an agent enters an infinite loop, observability is your only way to find the "why."
- **Task 5 Deliverable:** A docker-compose.yml that launches your entire Industrial AI system with one command.

